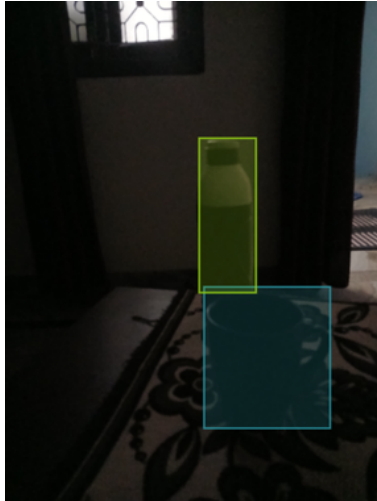
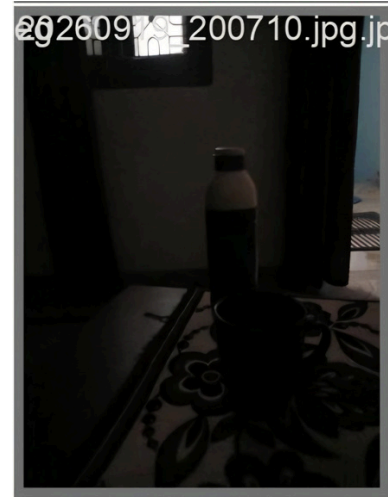


A4 - Failure Analysis

1) Low Light Scene: Both bottle and cup missed



Ground Truth



Model Output

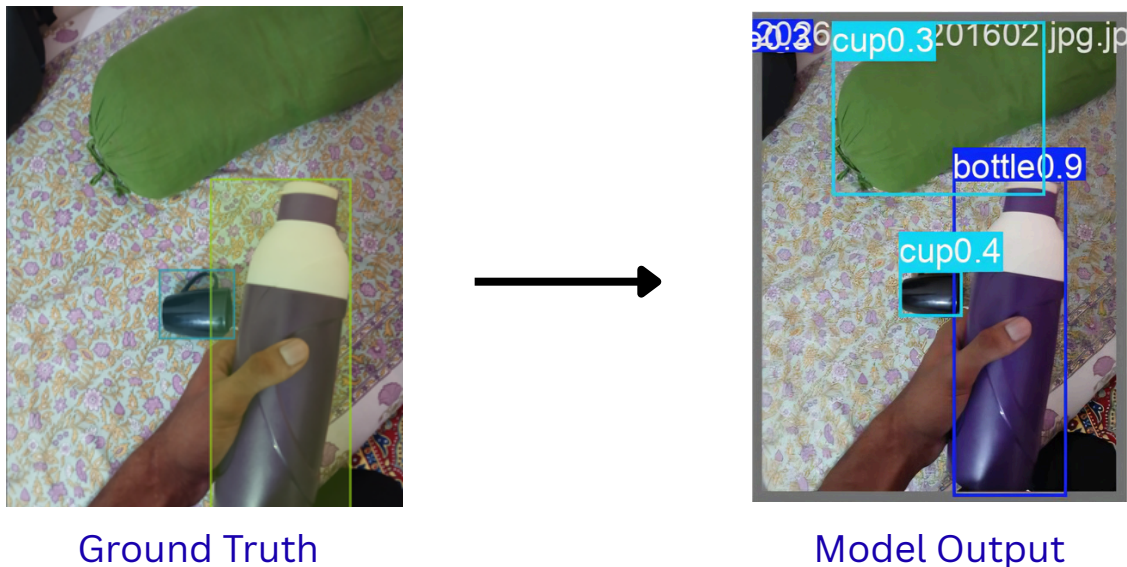
Observed prediction: The model neither detected the cup nor the bottle

Expected: Both cup and bottle should've been detected

Hypothesis: A total of 3 images were taken in such dark lighting conditions, out of which 2 ended up in the validation set, thus the model was only trained on only 1 image in such lights. now it could be said that during ultralytics' data augmentation, different exposure and brightness settings are used to tackle this situation, but the odds of such random augmentation to produce such dark results aren't very high, and besides both the bottle and the cup are dark in colour, the cup was infact black, there is almost no contrast, so it's really difficult for the model to detect the edges of the objects in such situation, does classify it altogether.

What i would change: Capture and train the model on more photos under such dynamic lighting conditions, introduce much more dynamic data augmentation during training varying the brightness, exposure and such features to wider ranges.

2) Background confusion: Classified pillow as a cup



Observed prediction: The model classified a pillow in the background as a cup

Expected: Only one cup and bottle were supposed to be detected in the photo

Hypothesis: Due to the curvy and rounded shape of the pillow, while classifying based on various edges of the pillow, the model must've took the pillow for a cup, as both are rounded in shape, also due to limited amount of negative-context data supplied to the model, the model doesn't seem very good at recognizing what rounded object is not a cup. The model has had very limited exposure to what similar looking objects are supposed to be explicitly rejected by it.

What i would change: Add more hard negative frames, containing similar looking objects to those of our target objects which are not really the target objects, thus training the model on what similar looking objects are not to be classified as our target objects, adding more background diversity in the dataset could also greatly help.

3) Horizontal overlap: Both bottle and cup missed when objects are horizontally stacked.



Ground Truth



Model Output

Observed prediction: Model failed to detect either of the bottle or the cup when the two were stacked horizontally.

Expected: It was supposed to classify and detect the two target objects.

Hypothesis: The training set had images where the bottle and cup were together vertically and in other poses, but failed to sufficiently cover the case of the two of them being stuck together horizontally, such a scenario couldn't have even been created using synthetic data augmentation, the model just did not seem to have expected for the two objects to be in such position because it did not see anything remotely similar during training, even the ground truth bounding boxes for this particular frame were greatly overlapping, the model did not receive such an image during its training.

What i would change: Capture the target objects in as many unique poses as i can, because synthetic augmentation could only do so much. While capturing the data i would vary the orientation, relative position and and the angles even more greatly for the model to be trained better.